EV DC Fast Charger – Complete Internal Components Explained in English

An EV DC Fast Charger is a system that converts AC electricity from the power grid into controlled DC power and supplies it directly to an electric vehicle's battery. It also communicates with the vehicle, monitors safety, and controls charging current and voltage.

Since you're preparing for an EV/automotive embedded systems role at AutoMech, focus on understanding the power electronics, charger controller, communication protocols, sensors, and protection circuits.

1. How a DC fast charger works



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The basic power flow is:

1. AC Grid Supply

Three-phase AC input

2. Input Protection & Filtering

Breaker, surge protection, EMI filter

3. AC–DC Rectifier / PFC

Converts AC to a regulated DC bus

4. DC–DC Converter

Adjusts output voltage and current

5. Output Contactor & Protection

Connects or isolates the DC output

6. EV Battery

Receives controlled DC power

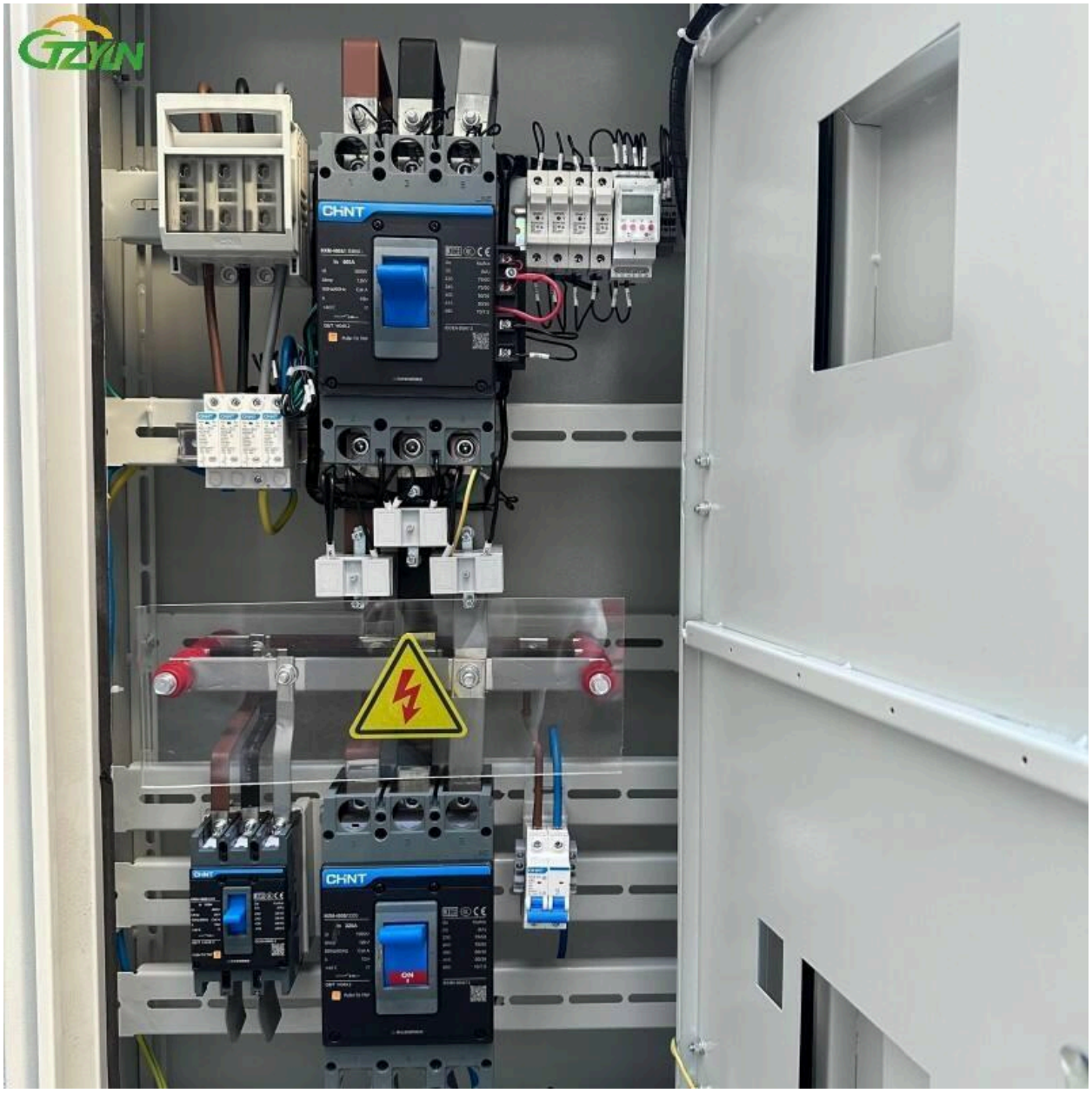
A separate control path operates alongside this power path:

Vehicle communication → Charger controller → Power-module commands → Sensors and feedback → Safety decisions.

The exact design varies by charger manufacturer, rated power, and charging standard.

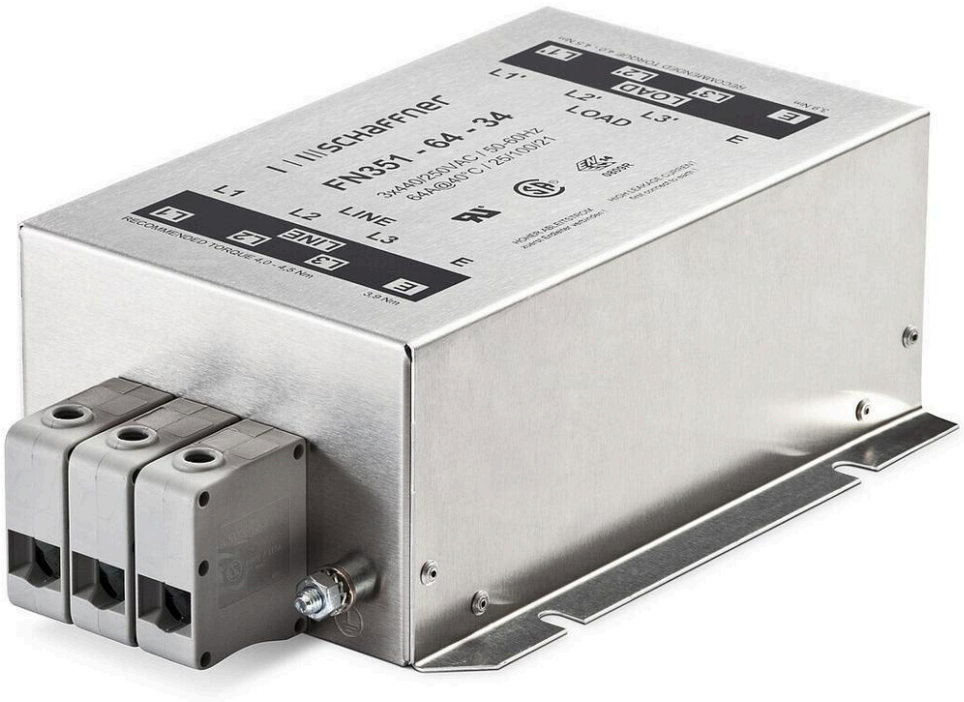
2. Complete internal components and their functions

A. AC input and protection section



1. Main circuit breaker (MCCB)

Protects the charger from excessive current and short circuits. It can disconnect the incoming electrical supply during a fault.



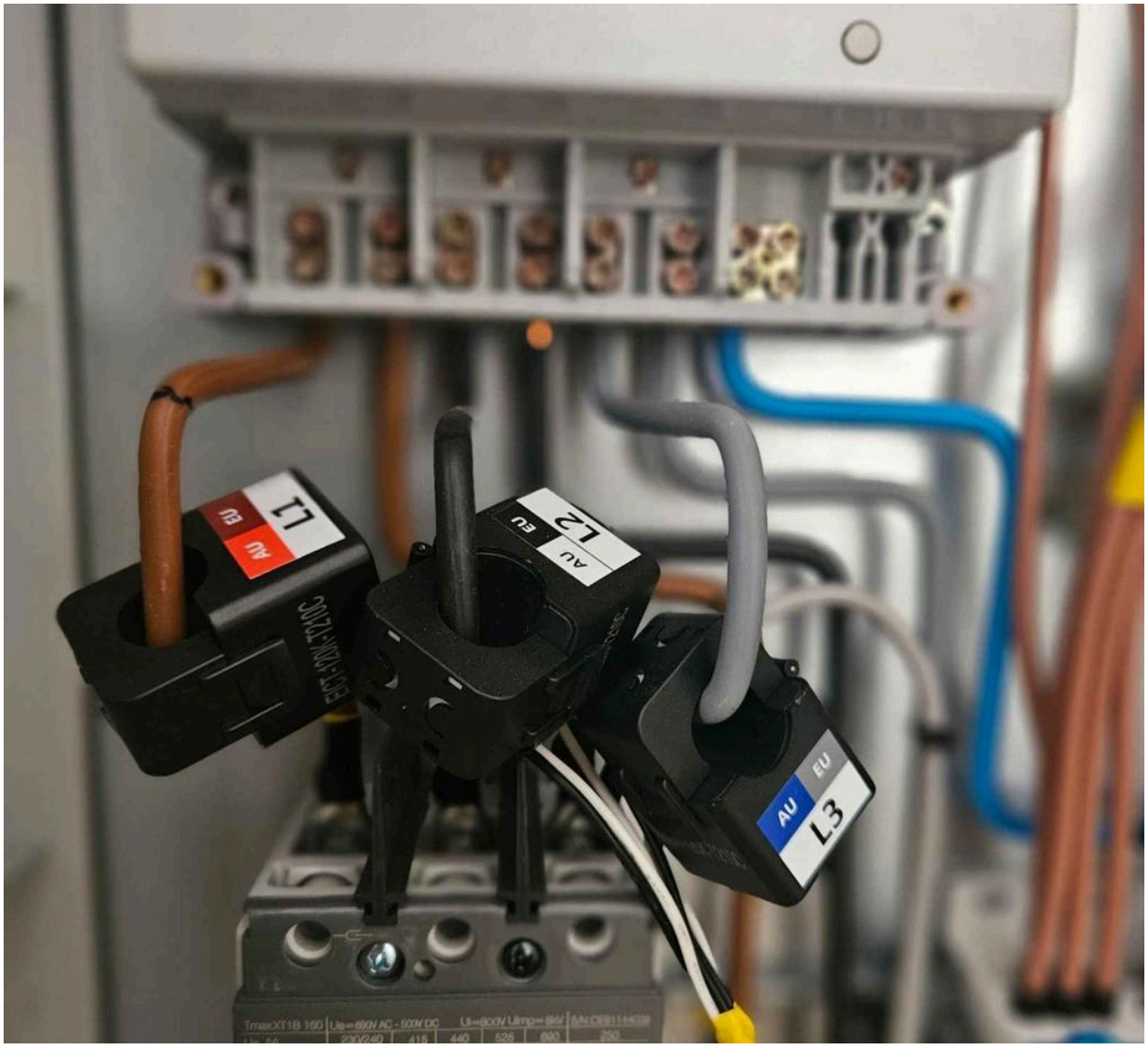
2. EMI filter

Reduces unwanted electrical noise generated by switching circuits and helps the charger comply with electromagnetic compatibility requirements.



3. Surge protection device (SPD)

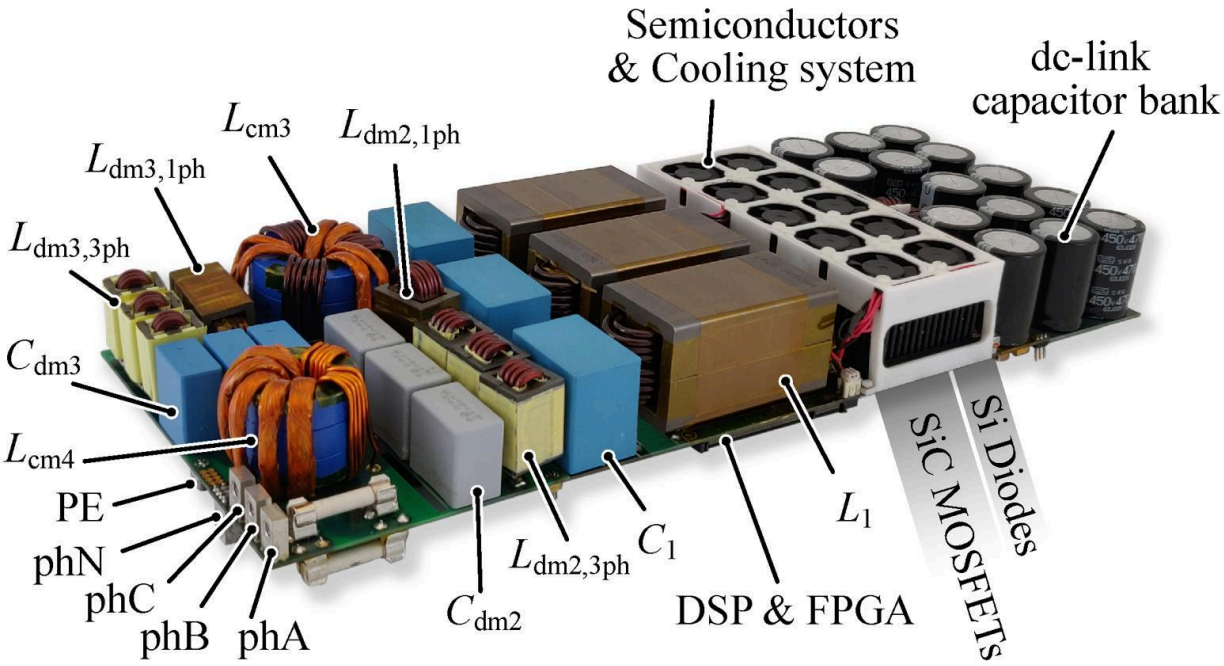
Helps protect electronics against transient voltage surges, such as those caused by lightning-related disturbances or switching events.



4. Voltage and current sensors

Measure incoming electrical conditions. The controller can use these measurements to detect abnormal conditions and manage power.

B. AC-to-DC conversion section



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5. Rectifier

The electrical grid supplies AC, but the charger needs DC for its power-conversion stages. The rectifier converts AC into DC.

6. Power Factor Correction (PFC) circuit

PFC controls the current drawn from the grid so that it follows the input voltage more closely. It improves power factor and reduces input-current harmonics.

7. Power semiconductor switches

These are the electronic switches used in conversion circuits.

- MOSFET: A voltage-controlled semiconductor switch used in many switching power converters.
- IGBT: A semiconductor switch commonly used in high-power conversion.
- SiC MOSFET: A silicon-carbide switch that can offer lower switching losses and high-temperature capability in suitable designs.

8. DC-link capacitors

These capacitors store electrical energy temporarily and smooth the DC voltage between conversion stages.

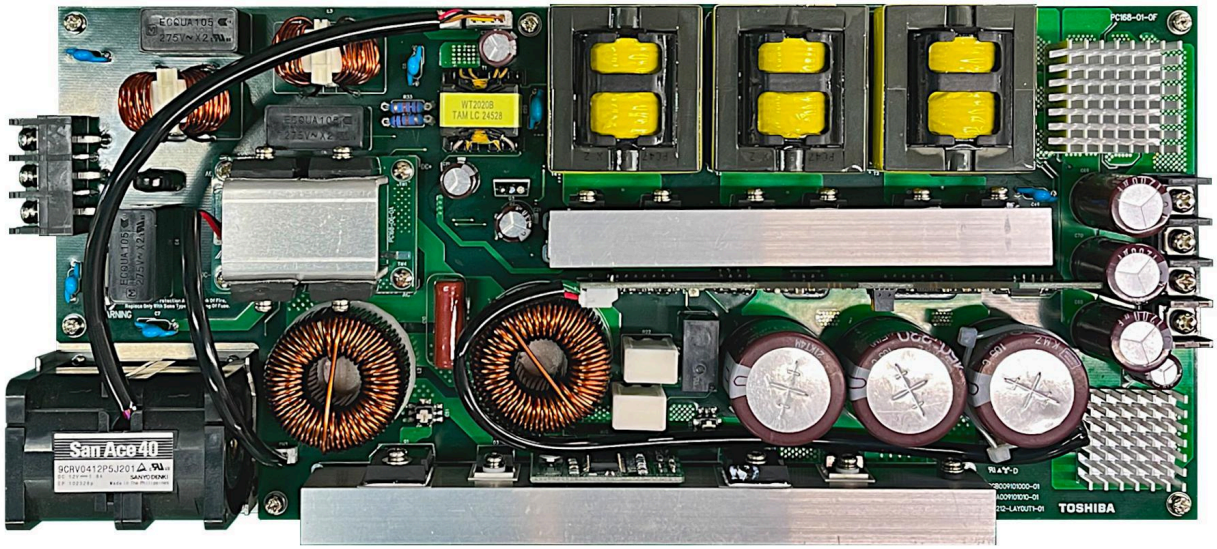
9. Inductors and transformers

- Inductors store energy in a magnetic field and help control current ripple.
- High-frequency transformers transfer energy and provide galvanic isolation in isolated converter designs.

The exact arrangement of the rectifier, PFC, DC bus and transformer depends on the charger architecture.

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C. DC-to-DC converter section



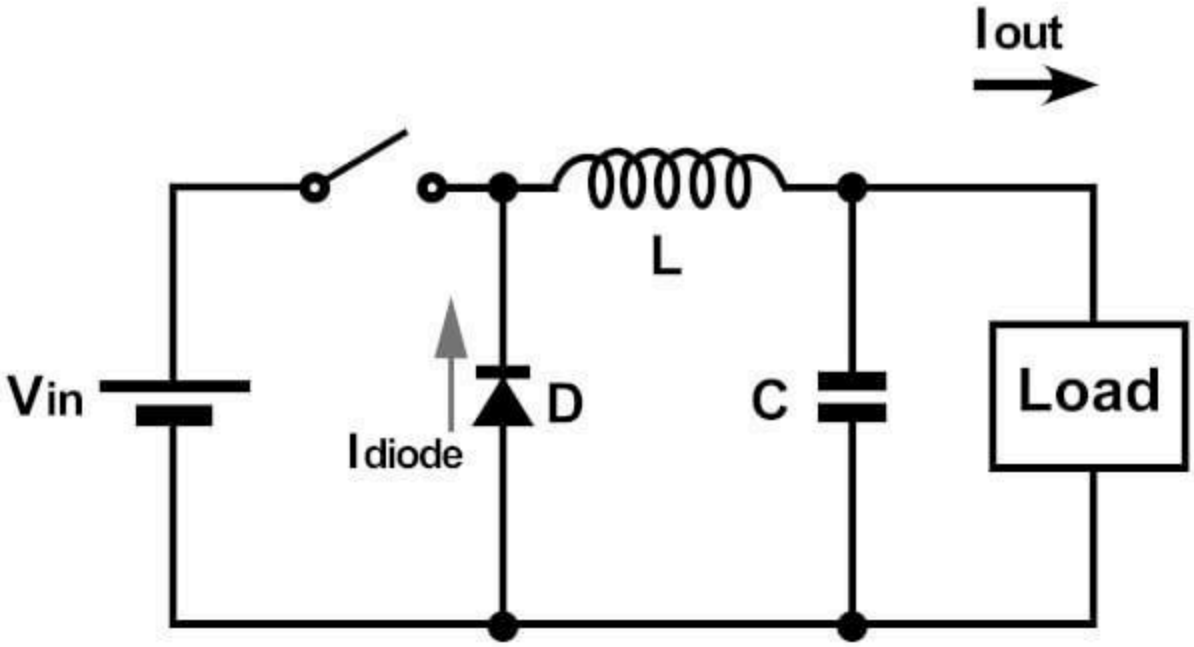
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10. DC–DC converter

This stage adjusts the DC voltage and current to meet the vehicle's charging requirements.

The two important converter types you should understand are:

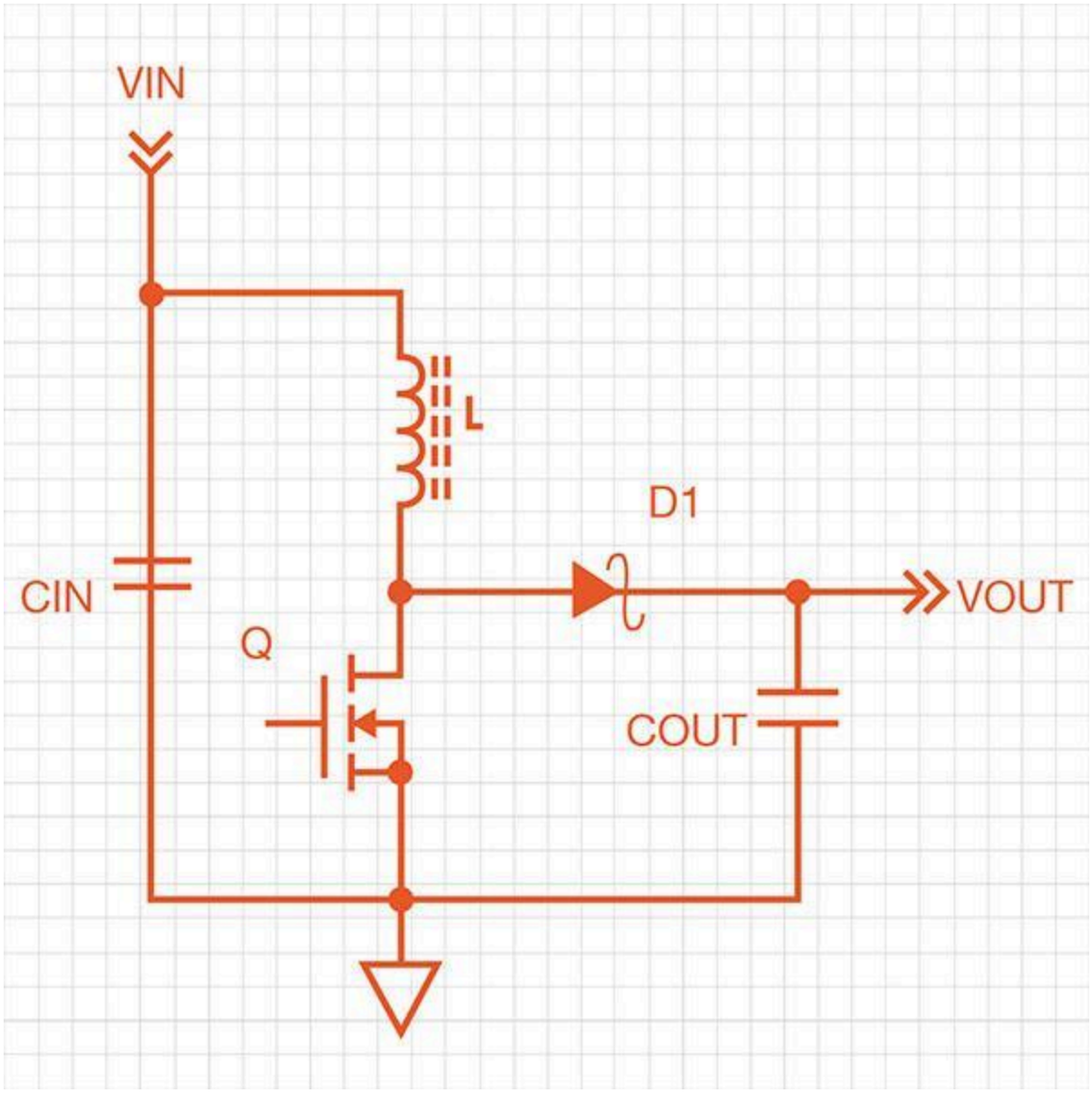
Buck converter



Purpose: Step down voltage.

Example: converting a higher DC bus voltage to a lower DC output voltage.

Boost converter



Purpose: Step up voltage.

Example: increasing a lower DC voltage to a higher DC bus voltage.

Buck–boost converter

It can increase or decrease voltage, depending on the topology and control strategy.

A DC fast charger does not necessarily use a separate buck–boost circuit. It may use an isolated LLC, phase-shifted full-bridge, dual-active-bridge, or another DC–DC topology.

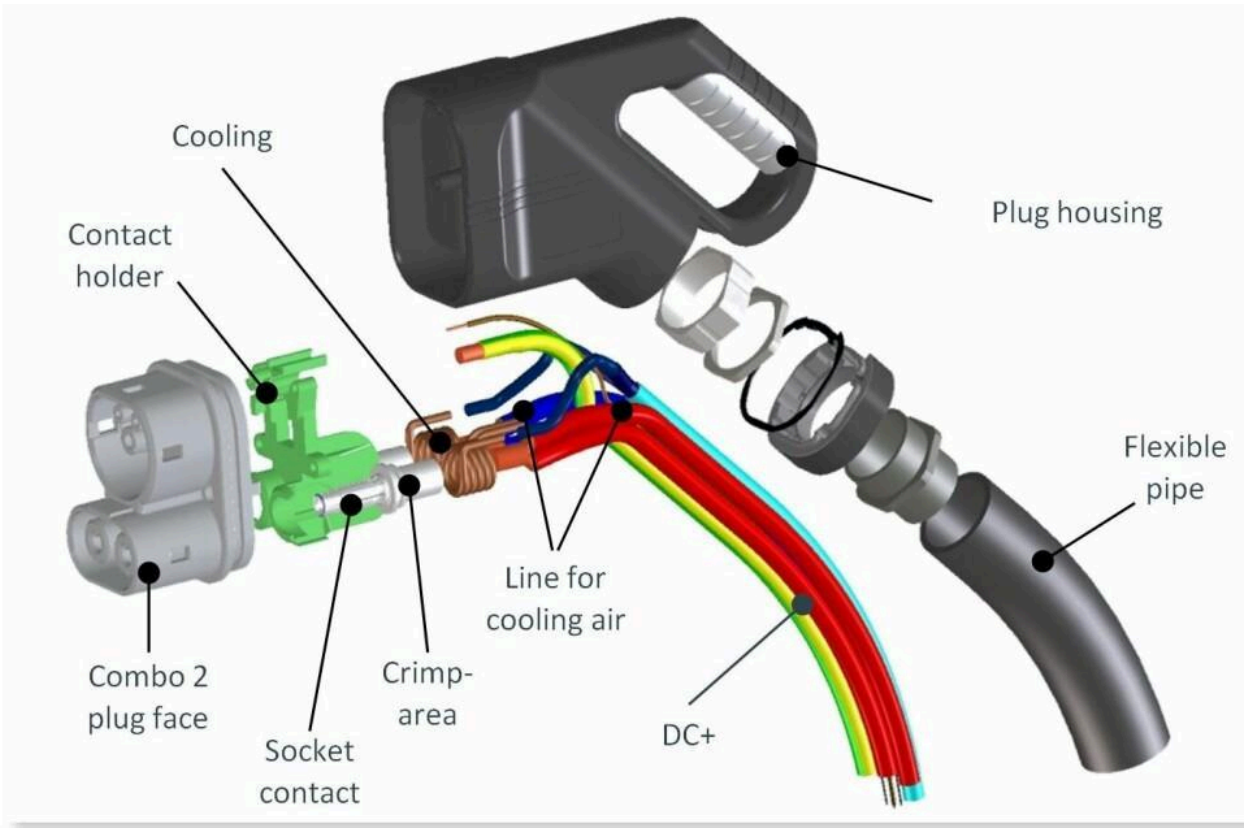
Interview point: The converter controls the charging output; the vehicle's battery-management system (BMS) communicates the battery's permissible charging conditions.

D. Output and battery connection section



11. High-voltage DC contactors

Electrically operated switches that connect or isolate the charging power path. The control system operates them only when the required safety conditions are met.



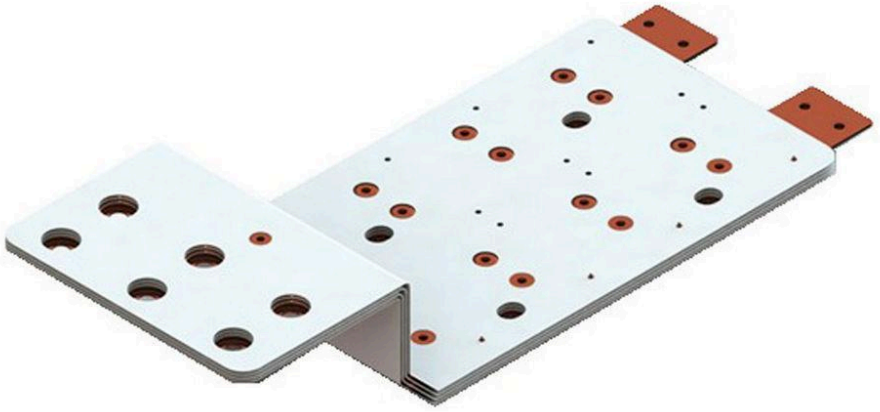
12. Charging connector and cable

Carry high-current DC power to the vehicle. The connector also contains communication and safety-related contacts as required by the charging standard.



13. Output voltage and current sensors

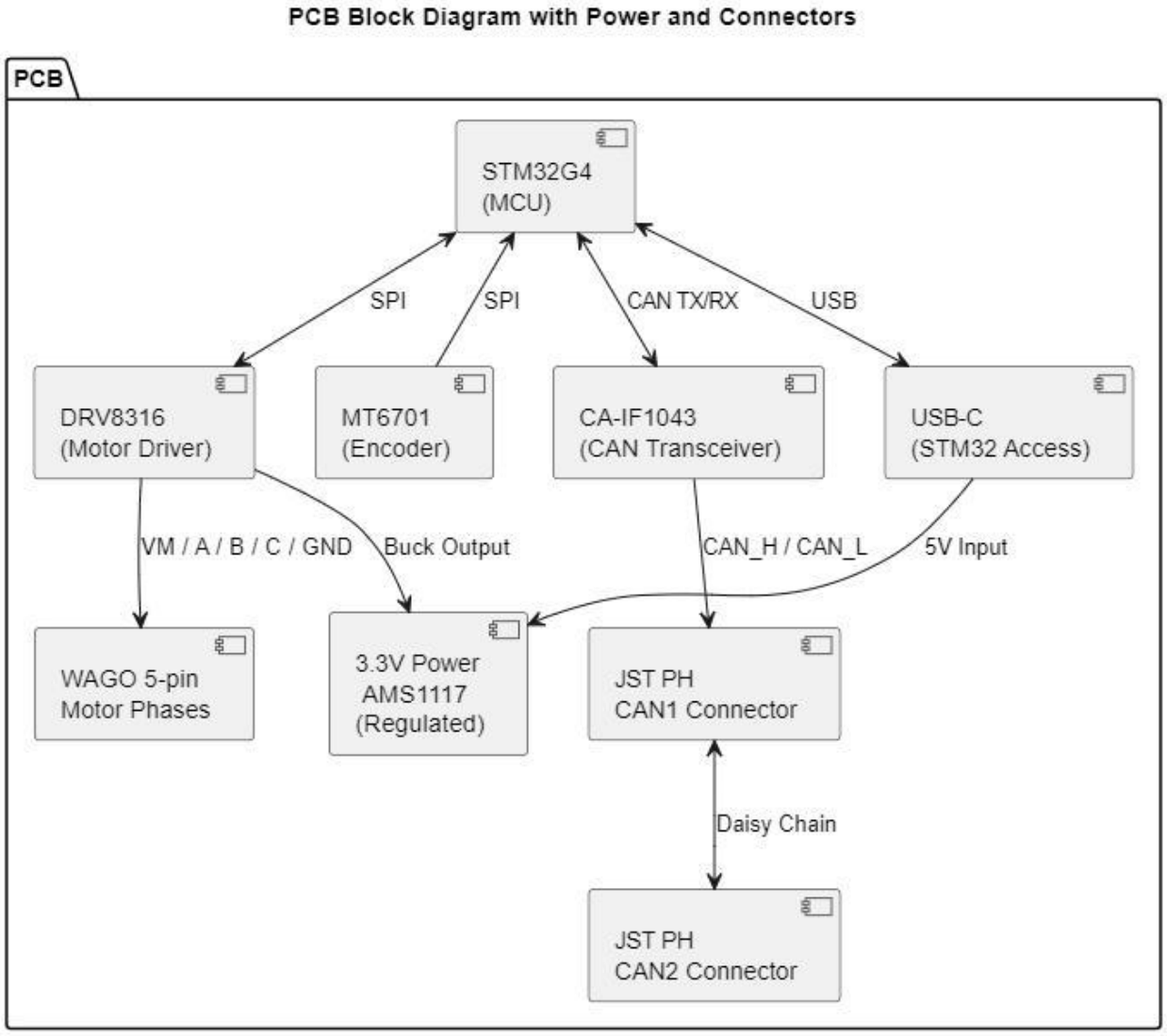
Measure the actual output supplied to the vehicle. Their feedback helps regulate charging and detect overcurrent or abnormal voltage.



14. Output capacitors, busbars and filters

Support stable power delivery, reduce ripple where required, and carry high current between power modules and the connector.

3. Charger controller and embedded electronics



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The charger is not only a power converter. It also needs embedded hardware and firmware to control the conversion process.

Component	Function
MCU (microcontroller)	Executes real-time control and monitoring firmware.

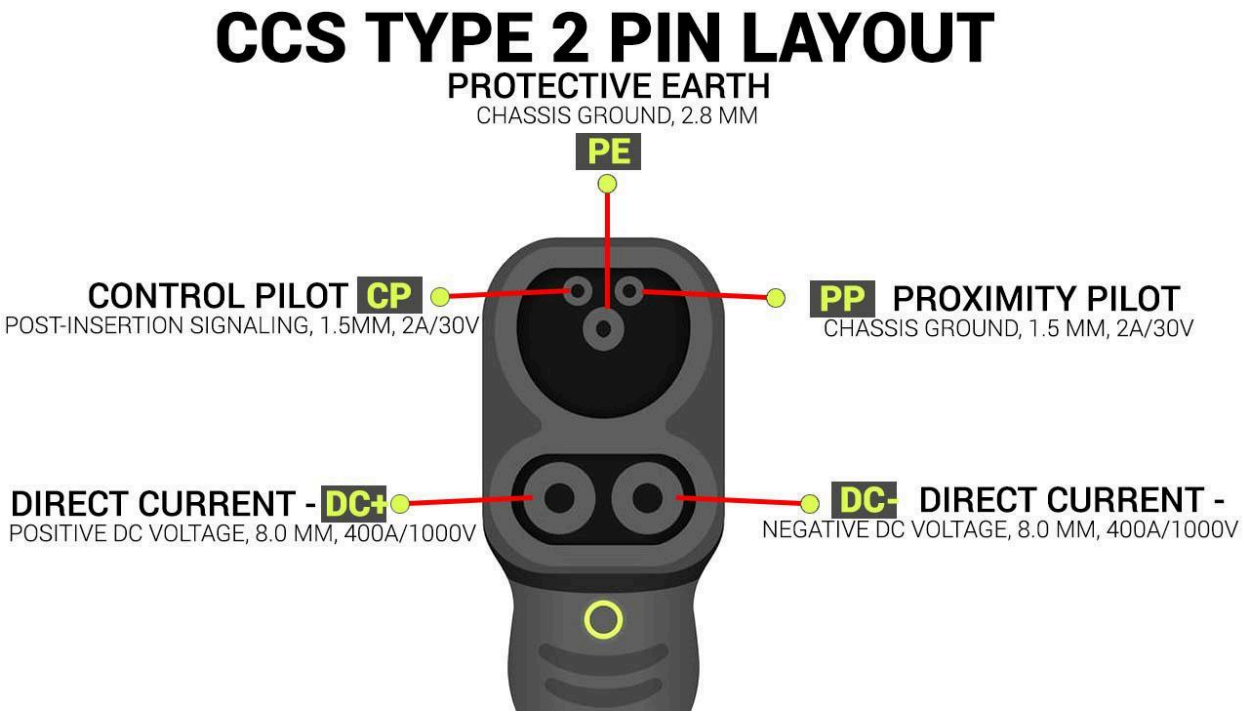
Gate driver	Converts control signals into suitable drive signals for power switches, often with isolation.
ADC (analog-to-digital converter)	Converts measured analog voltages into digital values for the controller.
CAN transceiver	Provides the physical electrical interface for a CAN communication bus.
Isolated power supply	Provides suitable low-voltage supplies for control and sensing circuits.
Watchdog circuit	Helps detect firmware that has stopped operating correctly.
EEPROM / Flash memory	Stores firmware, settings, and selected diagnostic information.
Temperature sensors	Monitor power modules, heatsinks, connectors, or other critical points.

For an embedded hardware internship, learn how these components connect on a schematic, how their supply rails are checked, and how to debug sensor, communication, and gate-driver faults. The MCU, sensing, gate drivers, auxiliary supplies and communication circuits form an interconnected control system.



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4. Communication between the charger and EV



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The charger and the vehicle must communicate before high-power charging begins.

Protocol / interface	Purpose
CCS2	Common DC charging connector system used in India and Europe.
DIN SPEC 70121	Defines digital communication for CCS charging.
ISO 15118	Defines EV-to-charger communication features, including supported smart-charging functions.
CAN	Used for communication between internal controllers or in some charging architectures.

Control Pilot (CP)	Provides a control and signalling interface, with details depending on the charging standard.
Ethernet / TCP/IP	Can carry higher-level charging communication, including supported ISO 15118 implementations.
OCPP	Communication between a charger and a charging station management system or backend.

Important: OCPP is generally for charger-to-backend communication, not the primary vehicle-to-charger charging protocol. CCS2 is the connector/system, while DIN SPEC 70121 and ISO 15118 define digital communication options.



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What information is exchanged?

The EV and charger may exchange or determine:

- Battery voltage and requested charging current.
- Permitted charging voltage and current limits.
- Charging status and progress.
- Readiness and safety conditions.
- Requests to increase, reduce or stop power.
- Fault or termination information.

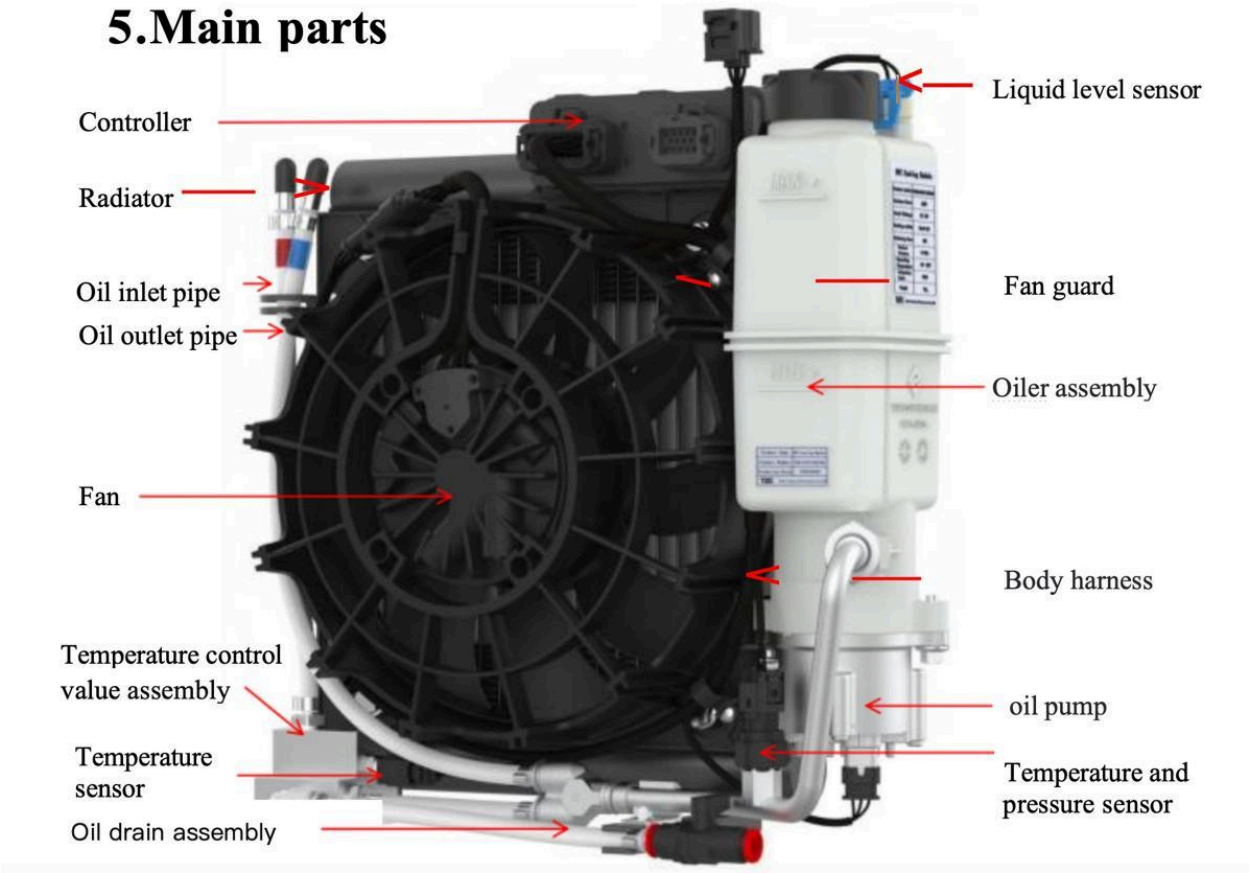
The exact messages and sequence depend on the communication standard and implementation.

5. Cooling system



15. Cooling fans and heatsinks

Remove heat from rectifiers, semiconductor switches and other power electronics.



16. Liquid cooling (where fitted)

Some high-power chargers use liquid-cooled power components or charging cables to manage heat.

The controller monitors temperature and can reduce output power or stop charging if a thermal limit is reached. Not every charger uses liquid cooling.

6. Safety and protection components

Component	What it protects against
Fuses	Excessive current and certain short-circuit faults.

Insulation monitoring device (IMD)	Insulation faults in applicable isolated DC systems.
Ground-fault / residual-current protection	Electrical leakage or fault conditions, as appropriate to the design.
Emergency stop	Allows charging to be stopped during an emergency.
Interlock and connector detection	Helps ensure the connector and system are in the required state.
Overvoltage and overcurrent protection	Abnormal electrical conditions.
Temperature monitoring	Overheating of power electronics and other monitored components.
Isolation barriers	Separate hazardous high-voltage circuits from low-voltage control electronics where required.

These protections work together. A fuse alone cannot protect against every electrical hazard, and the controller must not be the only layer of protection.

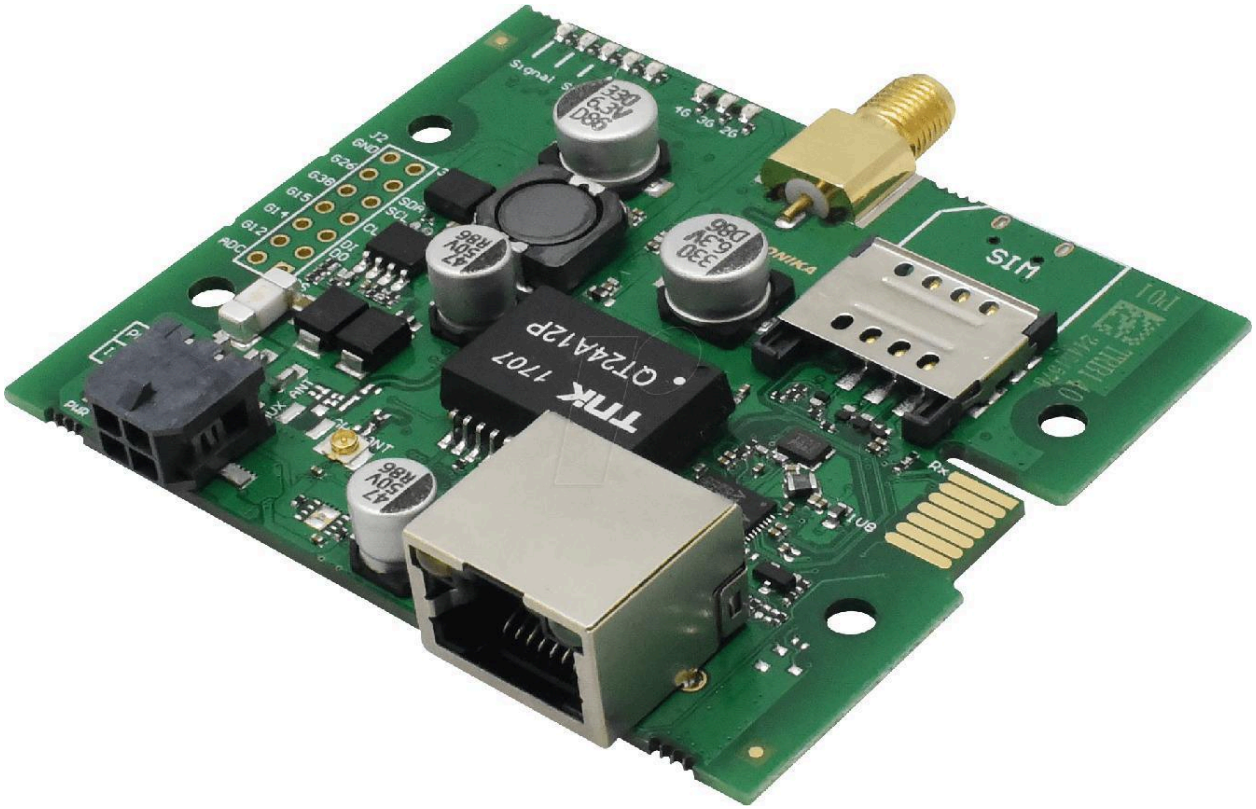
Safety note: DC fast chargers can contain lethal voltages and high stored energy even after the incoming supply is disconnected. Internal inspection and testing require appropriate training, isolation, and verification that the equipment is de-energized.

7. HMI and networking components



17. HMI (Human-Machine Interface)

The display and buttons allow users to start a session, view charging status and read error messages.



18. Network and payment interface

May include Ethernet, cellular connectivity, RFID readers, payment hardware and backend communication modules.



19. Energy meter

Measures delivered electrical energy, commonly in kilowatt-hours (kWh), for billing and monitoring.

8. What happens when you plug in an EV?

01

Connector detection

The charger detects the connection and checks the relevant signalling and safety conditions.

02

Communication

The charger and EV establish the supported communication session.

03

Parameter negotiation

They determine the permissible voltage, current and other charging limits.

04

Safety checks

The system checks required conditions, including relevant insulation, voltage and interlock checks.

05

Power delivery

The controller enables the power stage and operates the output contactors according to the design's safety sequence.

06

Closed-loop control

Sensors measure voltage, current and temperature. The controller adjusts power to meet the charging request.

07

Charging termination

When charging finishes or a fault occurs, power is reduced or stopped and the output is isolated according to the prescribed sequence.

9. Useful practical calculations

Suppose a charger delivers 150 kW to an EV at 500 V DC.

Using $P = V \times I$:

$$I = \frac{P}{V} = \frac{150{,}000}{500} = 300A$$

The output current is 300 A, assuming 150 kW is delivered at 500 V.

For another example, if the output voltage is 750 V at the same 150 kW:

$$I = \frac{150{,}000}{750} = 200A$$

This demonstrates why the voltage and current limits of both the charger and vehicle matter. Actual charging power is limited by the equipment ratings, the vehicle's requests, battery conditions and thermal limits.

10. What to study for an AutoMech interview

Focus on these topics in this order:

Priority 1 — Power electronics

AC–DC rectifier, PFC, DC link, buck and boost converters, MOSFETs, IGBTs, gate drivers, capacitors and inductors.

Priority 2 — Embedded hardware

MCU, ADC, PWM, voltage/current sensing, isolated power supplies, CAN transceivers and PCB debugging.

Priority 3 — Communication and control

CCS2, ISO 15118, charging sequence, contactor control, fault handling and closed-loop control.

Priority 4 — Schematic and PCB testing

Read a reference schematic, identify power rails, check components, use a multimeter safely, and understand DFM/DFT and EMI/EMC.

Free technical resources

- onsemi — DC EV Charger interactive block diagram
- NITI Aayog — Fundamentals of Electric Vehicle Charging Technology and Grid Integration (PDF)
- Nexperia — DC EV charging station components and design resources

Your next practical step: Learn to draw the complete charger block diagram from memory, then study the rectifier/PFC stage, the DC–DC converter, and the embedded controller schematic one by one. These are the foundations for understanding EV charger controller and embedded hardware roles.